

JAY CONSEJO

Full-Stack Builder & AI-First Founder

Project Portfolio | April 2026

Four products built across web, AI, and Web3 — from an AI-powered news platform serving daily personalized digests, to a video production studio automating YouTube workflows, to an on-chain campaign verification system on Base. Each project was architected, built, and deployed as a solo founder with AI-assisted development.

Murmura Studio AI VIDEO PRODUCTION

AI-powered video production platform for YouTube creators. Automates the full content pipeline — research, scripting, voiceover, rendering, and publishing.

FRAMEWORK	Next.js 16 (App Router, Turbopack)	FRONTEND	React 19, TypeScript 5, Tailwind CSS 4
AI / LLM	Vercel AI SDK 6, Anthropic, Google Gemini	VIDEO	Remotion 4 (programmatic rendering)
VOICE	ElevenLabs, Edge TTS, Google Cloud TTS	AUDIO	Deepgram (speech processing)
DATABASE	PostgreSQL (Railway) + Prisma 7 ORM	STORAGE	Cloudflare R2 (S3-compatible)
AUTH	NextAuth.js 5 (Google OAuth)	DEPLOY	Vercel (serverless, 5-300s functions)

KEY ARCHITECTURE DECISIONS

- Server Components for data-heavy pages, client components only where interactivity requires it
- Vercel AI SDK streamText + tool-calling for autonomous content workflows
- Cloudflare R2 over S3 — zero egress costs for media-heavy platform
- Cost-optimized AI cascade: free tiers first, paid only on overflow

Provely WEB3 CAMPAIGN PLATFORM

On-chain campaign verification platform on Base. Brands create campaigns, creators submit proof of social content, and rewards are distributed via smart contracts with cryptographic verification.

FRAMEWORK	Next.js 14 (App Router)	FRONTEND	React 18, TypeScript, Tailwind CSS 3
API LAYER	tRPC 11 (end-to-end type-safe RPCs)	WEB3	Wagmi 2, Viem 2, RainbowKit 2
AUTH	SIWE (Sign-In with Ethereum)	CONTRACTS	Solidity 0.8.27, Hardhat, OpenZeppelin v5
CHAIN	Base (mainnet + Sepolia testnet)	DATABASE	PostgreSQL (Railway) + Prisma 7 ORM
CACHE	Upstash Redis (rate limiting)	DEPLOY	Vercel + Base (on-chain)

SMART CONTRACTS

- **CampaignEscrow** — ERC20 deposits, ECDSA-verified claims, reentrancy protection
- **CampaignBadge** — ERC1155 achievement badges with IPFS metadata
- Deployed on Base mainnet and Sepolia testnet via Hardhat

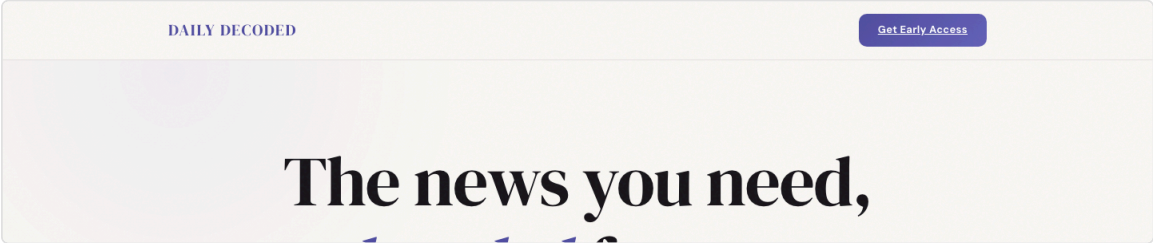
KEY ARCHITECTURE DECISIONS

- tRPC for fully type-safe API — no REST schema drift between frontend and backend
- SIWE over traditional auth — wallet-native identity, no password management
- ECDSA signature verification on-chain — server signs, contract verifies (gas-efficient)
- Base chain for low fees and Coinbase ecosystem alignment

Decoded Daily News

AI NEWS PLATFORM

AI-personalized daily news digest for professionals. Fetches from 40+ sources, uses multi-step AI summarization, and delivers role-personalized digests via email and web at 6 AM daily.



BACKEND	Python FastAPI (async-first)	FRONTEND	HTML/CSS/JS, Tailwind CSS (CDN)
AI / LLM	Gemini, Vertex AI, Groq, Claude Haiku	DATABASE	Supabase (PostgreSQL + Auth + RLS)
SCHEDULER	APScheduler (in-process cron)	EMAIL	SendGrid (HTML + PDF attachments)
SCRAPING	RSS + NewsData.io + Trafilatura	PDF	ReportLab (branded digest PDFs)
BACKEND HOST	Railway PaaS (staging + production)	FRONTEND HOST	Cloudflare Pages (global CDN)

- KEY ARCHITECTURE DECISIONS
- Supabase Row-Level Security — frontend uses anon key safely, all access controlled at DB level
 - Hash-pinned dependencies (SHA-256) after supply chain incident with litellm package
 - 2-step legal-safe AI summarization — max 3 consecutive source words, original analysis only
 - AI provider cascade: Gemini free → Vertex AI → Groq → Claude Haiku (cost optimization)
 - Cloudflare Pages for frontend — zero cost, global CDN, automatic HTTPS

jayconsejo.com

PORTFOLIO & AGENCY SITE

Personal portfolio and AI agency landing page. First project built entirely through the HADP agency pipeline from design through deployment.

FRAMEWORK	Next.js 16 (App Router, Turbopack)	FRONTEND	React 19, TypeScript 5, Tailwind CSS 4
UI	shadcn/ui + Base UI React, Lucide icons	DEPLOY	Vercel (automatic Git deploys)

- KEY ARCHITECTURE DECISIONS
- Design-neutral skeleton — structural foundation reskinable via CSS variables
 - Full SEO infrastructure: Metadata API (OG/Twitter), sitemap.ts, robots.ts
 - 6 security headers: HSTS, X-Frame-Options, nosniff, referrer, permissions policy

Hybrid AI Dev Pipeline (HADP)

AI AGENCY ORCHESTRATION

Self-hosted AI-powered software development agency. 19 specialized agents across 3 teams (Dev, Marketing, Creative) orchestrated via Slack, n8n, and Claude — managed by a non-technical CEO.

ORCHESTRATION	n8n (Docker on Mac)	SENIOR AGENT	Claude Code CLI (Opus)
LOCAL MODELS	Ollama — Qwen 2.5 14B, Gemma 4 26B	NETWORKING	Cloudflare Tunnel + Access
COMMS	Slack (bot + webhooks)	HARDWARE	Mac M3 + PC RTX 5070 (ethernet)

- KEY ARCHITECTURE DECISIONS
- Hybrid cloud: Mac for orchestration + PC GPU for local inference — zero API cost for simple tasks
 - Chief of Staff pattern — single brain (Claude Opus) interprets all CEO requests before delegation
 - Security-first: bearer token auth, 127.0.0.1 binding, rate limiting, secret stripping, path whitelisting
 - Bridge decomposed from monolith to 10 modules for maintainability and token efficiency